

Human Robot Interaction

Introduction to Social Robot

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Department of Computer Science

| Defining the HRI

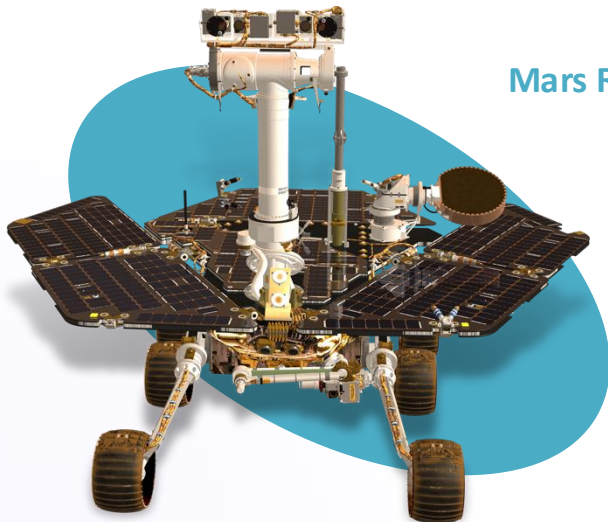
Human-robot interaction (HRI) is the interdisciplinary study of dynamics interaction between humans and robots. Interaction requires communication between robots and humans, that may take several forms.

Remote interaction

The human and the robot are not co-located and are separated spatially or even temporally.

Proximate interaction

The humans and the robots are co-located



Mars Rover



Social robots

Defining the Social Robot

More Than Just Automation

A robot is defined in the **International Standard of Organization**¹ as a reprogrammable, multifunctional manipulator designed to move material, parts, tools or specialized devices through variable programmed motions for performance of a variety of tasks.

As a subset of robots, **social robots** perform any (or all) of these processes in the context of a social interaction.

A social robot is an autonomous agent that interacts and communicates with humans by following expected social behaviors and rules.

Autonomy

Can function without continuous human control.

Social Interface

Uses cues like gaze, gestures, and speech.

Adaptability

Learns from interaction to improve communication.



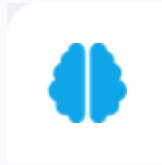
¹ It is an independent international organization that develops and publishes globally recognized technical standards

| Core Components of Social Robotics



Embodiment

The physical form is crucial. Unlike screen-based agents, a robot shares our physical space, allowing for non-verbal communication like proximity (proxemics) and touch.



Social Cognition

The ability to perceive, interpret, and respond to human intentions and emotions. This involves "Theory of Mind": understanding that others have beliefs and desires.



Perception System

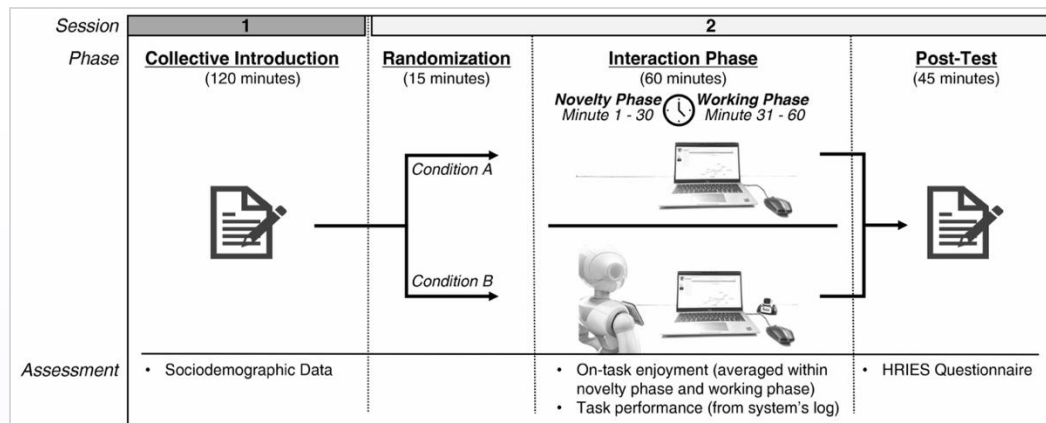
Advanced sensors (cameras, microphones, tactile sensors) that allow the robot to track faces, recognize speech, and detect emotional states in real-time.

Core Components of Social Robotics

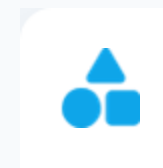
The Power of Embodiment

Why build a robot when you have a tablet?

Research shows that physically embodied agents elicit stronger social presence and engagement than on-screen avatars [1].



The study consisted of two 2h sessions. Participants in condition (A) were learning with an ITS with virtual robotic agents. In the condition (B), participants were additionally accompanied by physically present robotic agents. The experimental setup included a robotic peer agent (Cozmo robot) and a robotic tutor agent (Pepper robot).



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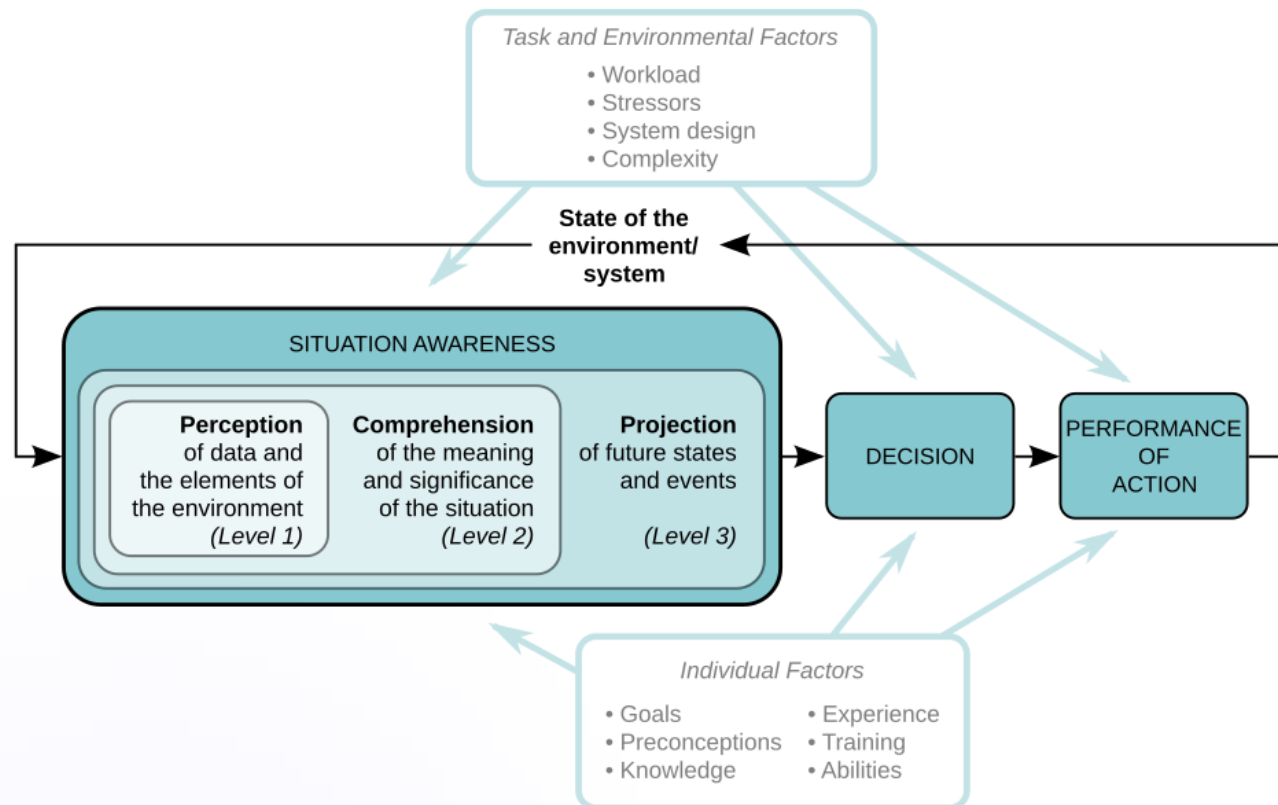
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[Source Image] Endsley, M. R. (1995). Measurement of situation awareness in dynamic systems. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 37(1), 65–84. <https://doi.org/10.1518/001872095779049499>

| Applications



Elderly Care

| Applications



Elderly Care



Autism Therapy

| Applications



Elderly Care



Autism Therapy



Education

Applications



Elderly Care



Autism Therapy



Education



Service

Applications



Elderly Care



Autism Therapy



Education



Service



Art

| Applications



An **artistic installation** is a spatial, immersive form of art designed to transform the environment rather than presenting a single object. It often integrates multiple sensory stimuli (visual, auditory, tactile) so the audience experiences more than one sense at a time.

Museums and galleries use installations to enrich visitor engagement, moving beyond passive observation toward active or interactive participation. Successful installations create emotionally or psychologically meaningful experiences, sometimes aiming for well-being, introspection, or emotional balance.

(Another viewpoint: not every installation aims at well-being; many provoke.)

| L'importanza della UX nelle Installation Artwoks

Le opere d'arte installative (a volte descritte anche come "ambienti")

- occupano spesso un'intera stanza o uno spazio espositivo attraverso cui lo spettatore deve camminare per poter interagire pienamente con l'opera.
- Alcune installazioni, tuttavia, sono progettate semplicemente per essere osservate passando intorno ad esse e contemplate, oppure sono così fragili che possono essere viste solo da una porta o da un'estremità della stanza.
- Ciò che rende l'arte installativa diversa dalla scultura o da altre forme d'arte tradizionali è che **essa rappresenta un'esperienza completa e unificata**, invece di una mostra di opere separate e individuali. L'attenzione rivolta a come il visitatore vive l'opera e il desiderio di offrirgli un'esperienza intensa costituiscono un tema dominante nell'arte installativa:

"The main actor in the total installation, the main centre toward which everything is addressed, for which everything is intended, is the viewer." (artist Ilya Kabakov)



Interactive Public Sculpture: The Pool

<https://www.youtube.com/watch?v=TJf2qfrtAcA>

Applications

Robotics as an Artistic Medium

Robots are increasingly used in art installations to explore themes of identity and digital reflection.

Example 1



"Can't Help Myself"

Sun Yuan and Peng Yu [2016]

Example 2



"Dynamics of a Dog on a Leash"

Takayuki Todo

Example 3



"HitchBOT" [2014]

| Robotics as an Artistic Medium



"Can't Help Myself" by *Sun Yuan* and *Peng Yu* [2016]

What It Is

It consists of a large, industrial robotic arm placed inside a clear acrylic box, like a cage or a laboratory observation room. On the floor is a pool of a red liquid.

What It Does

The robot's job is to continually sweep the red liquid back towards the center of the box using a spatula. However, the viscous liquid constantly goes away from the center. The robot has visual sensors that detect when the fluid crosses a boundary, triggering the arm to start its frantic sweeping motion again.

What It Means (Interpretation)

The piece is a powerful and dark metaphor for several modern issues: The robot is engaged in a pointless task that must be repeated forever. It symbolizes the continuous, exhausting labor of modern life that leads nowhere. The artists also programmed the robot to perform random, almost humorous "dance moves" when its task is momentarily finished. This humanization makes the machine even more "sad".

Robotics as an Artistic Medium



“Dynamics of a Dog on a Leash” by *Takayuki Todo*

What It Represents

The chain represents the need for strict safety rules and ethical limits on Artificial Intelligence (AI). The art warns that without control, powerful AI could become a threat. The artwork taps into the human fear that we might lose control over the intelligent machines we create. The violent, desperate struggle of the chained machine often makes viewers feel pity for it, even though they know the robot is just code and metal. This questions how we feel about non-living things.

What is it

It is an art installation that uses a real, commercially available robot dog (like the ones used in industry or security). The robot dog is secured to the floor with a heavy chain.

What it does

The machine is specially programmed to be aggressive. It Tries to Attack: When people walk near the robot, its sensors detect them. The robot dog then starts to be violent as if it is trying to attack the visitors. A metal chain physically stops the robot from reaching anyone.

Robotics as an Artistic Medium



“HitchBOT” [2014]

The Message

It was designed to test whether humans would trust and care for a vulnerable, non-human companion. The experiment ended sadly in the U.S. when it was found destroyed and decapitated in Philadelphia, highlighting the darker side of human interaction and lack of empathy for machines.

What it Is

HitchBOT was a harmless robot, built mostly out of household objects (like a bucket, pool noodle arms, and rubber boots). It had a digital face and could communicate with people using AI.

What it did

The robot's goal was to hitchhike across Canada, Germany, and the Netherlands. It could not move by itself; it relied entirely on the kindness of strangers to pick it up, carry it, and charge its battery.

| The Furhat Platform

Back-Projected Face Technology

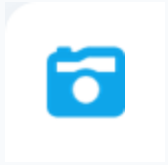
Unlike traditional robots with fixed skins, Furhat uses a semi-transparent facial mask with a micro-projector inside.

This allows for **instant morphing** of the robot's identity. It can change age, gender, and appearance in milliseconds while maintaining high-fidelity facial animation.



Our Innovation

Face maker Tool



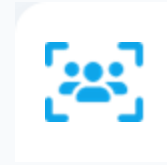
Capture

We have developed a custom pipeline that allows us to take a standard photo of **any person's face** (yours, a celebrity, or someone familiar).



Map & Project

The tool automatically maps the 2D image onto the 3D Furhat mask geometry, aligning eyes, mouth, and key facial features.



Experience

The result: The robot physically embodies the person. It creates a sensation of meeting a digital twin in the real world.

Our Innovation

Face maker Tool



Capture



Map & Project



Experience

CRpF - Customized Robot projection Face: Craft Your Digital Twin

Craft Your Digital Twin
Bring your identity to Furhat

Character Name

Enter your name

Choose your eyes

16 eye color options (2 rows of 8).

Upload your frontal face photo

Drag and drop your photo here

[or browse files](#)

Create Character

Our Innovation

Face maker Tool



Capture



Map & Project



Experience

1

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Character Name

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2 →

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Capture



Map & Project



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Enter your name

2 → **Choose your eyes**

3 → **Upload your frontal face photo**
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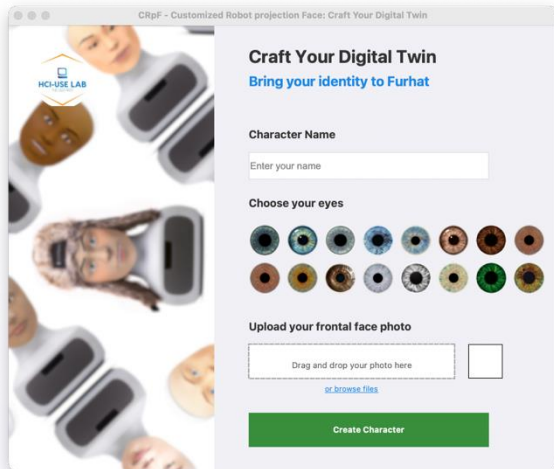
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Create Character



Personalization

Face maker Tool



| Scenario

Le opere viste sono tutte accomunate dall'esplorazione del confronto tra uomo e macchina, tra libertà e controllo, o tra vulnerabilità ed empatia.



La Challenge

Progettare il concept di un'installazione artistica interattiva in cui il robot sociale **Furhat**, dotato di un volto personalizzato, contribuisce alla creazione di un'esperienza immersiva centrata su un tema legato all'identità, alla percezione di sé o alla relazione umano–macchina.

Tema suggerito: *La consapevolezza di sé*

Come cambia la percezione che abbiamo di noi stessi quando un'entità artificiale ci osserva, dialoga con noi e riflette — in modo più o meno fedele — aspetti della nostra identità?

Temi alternativi :

- ***Empatia artificiale***
- ***La molteplicità dei sé (fisico, digitale, emotivo)***
- ***La memoria come costruzione***
- ***L'altro che diventa specchio***



Task da svolgere

1. Ideare lo scenario dell'installazione

- Dove si trova? Come appare lo spazio? Come viene guidato il visitatore?
- Che ruolo ha Furhat? È un narratore, uno specchio, un confidente, un provocatore?

2. Progettare l'esperienza utente (UX)

- Quali sono le fasi dell'interazione?
- Il visitatore deve parlare? Muoversi? Rispondere a stimoli visivi/sonori?
- Quali feedback fornisce il robot (voce, espressioni, proiezioni)?

3. Personalizzare il volto di Furhat con un volto familiare (il proprio o quello di una persona a voi vicina)

- Cosa comunica il design rispetto al tema scelto?

4. Definire un messaggio chiaro

- Cosa deve portare con sé il visitatore dopo l'esperienza?
- In che modo l'installazione lavora sul concetto di consapevolezza, identità o relazione?

5. Riflettere sui principi di HCI e usabilità

- Quali accorgimenti rendono l'interazione fluida, inclusiva e non invasiva?
- Come evitare uncanny valley* o discomfort non intenzionali?
- Quali problemi etici emergono (privacy, proiezione di identità, risposta emotiva)?

* la reazione di disagio che le persone provano verso robot o simulazioni umane che sono quasi, ma non del tutto, realistiche.



Consegnare

- **Descrizione del concetto e dell'idea di installazione, rispetto al tema scelto**
- **Wireframe (o storyboard) dell'interazione**
- **Descrizione del volto personalizzato e del suo significato**
- **Mappa dell'esperienza utente**
- **Pitch finale (3 minuti) con Powerpoint**